LongRange-PFN: KL-Distilled Recursive Latent Summaries for Prior-Fitted Networks beyond the Context Window

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Abstract

Prior-fitted networks (PFNs) predict a Bayesian posterior in one forward pass, but only up to the context size seen in training: attention is quadratic, and quality degrades past that window. LongRange-PFN compresses a distant part of the context, D', into k latent tokens $z = \mathrm{rep}(D')$ with a learned summarizer, and predicts from the remaining local context D with z added as extra tokens. A student is distilled from a frozen, full-context teacher PFN by matching predictive distributions; since both use the same discretized output, the KL between them is exact. A recursive summarizer, $z_t = \mathrm{rep}(C_t, z_{t-1})$, folds the distant context one chunk at a time and keeps constant state, so inference is no longer capped by the training window. On Gaussian process priors, where the true posterior is known, the student's NLL falls 0.03–1.30 nats below a lower bound available to any method that sees only D, closing 97% of the truncation gap, and in a streaming setting with 4× the training window the recursive summary also beats a teacher restricted to the most recent points, recovering ≈ 39% of the remaining headroom. Two things do not work yet: multi-chunk summarization stays noticeably worse than single-shot even after fixing the model-selection criterion that first hid it, consistent with error compounding across recursive steps; and bounded-state inference still loses to classical full-context baselines on this prior, where exact inference is cheap. Against the training-free selectors studied by CRUMB, at a matched raw-token budget, the summary is closest to the full-context teacher at $n = 16$; per-query nearest neighbors win at $n = 8$; and under a disjoint covariate shift the selectors degrade less than a summary that was trained only on exchangeable data.

1 Introduction

Prior-fitted networks (Müller et al., 2022) treat Bayesian prediction as in-context learning: pre-train a transformer on data sets drawn from a prior, then at test time feed it a training set D_{tot} and a query x and read off an approximation to $p(y \mid x, D_{\mathrm{tot}})$. TabPFN (Hollmann et al., 2023, 2025) turned this into a working tabular model, but the context is the model's memory: every observation is a token, attention is quadratic, and the network has never seen contexts larger than some N during training. Past that size it is answering a question it never practiced.

Three lines of work address this (Section 4): optimizing a compact context per dataset at test time (Feuer et al., 2024; Ma et al., 2024), retrieving real neighbors instead of the whole context (Thomas et al., 2024), and, in language models, compressing prompts into soft tokens (Mu et al., 2023; Ge et al., 2024; Chevalier et al., 2023). None gives a PFN a fixed-size, reusable summary of what the far-away data contributed to the posterior.

LongRange-PFN splits $D_{\text{tot}} = D' \cup D$, compresses D' into k latent tokens $z = \text{rep}(D')$ with a learned summarizer (one forward pass, no test-time optimization), and predicts from $[\, z \mid D \mid x \,]$ with a standard PFN backbone that treats z as ordinary context. Training distills a frozen, full-context teacher PFN: teacher and student share the same discretized ("bar") output, so the KL between their predictions is an exact categorical KL with no sampling.

The summarizer is also recursive: $z_t = \text{rep}(C_t, z_{t-1})$ folds a stream of context chunks into a fixed-size summary, mirroring the Bayesian filtering update $p(\rho \mid C_{1:t}) \propto p(C_t \mid \rho)\, p(\rho \mid C_{1:t-1})$ over a latent task ρ. This decouples how much data the system uses from the window any component was trained with: distilled only on contexts up to 80 points and 3 chunks, at inference the recursion handles 320 points in five chunks, and beats a strong recency baseline rather than just matching it (Table 3).

Contributions.

1. A summary-conditioned PFN trained by predictive KL distillation (Section 3): $\text{rep}_\phi(D')$ is learned end-to-end so that $q_\phi(y \mid x, D, z)$ matches a frozen teacher's $p_\theta(y \mid x, D_{\text{tot}})$, with an exact bucket-level KL.

2. A recursive summary update (Section 3.3) that gives bounded-state, anytime PFN inference beyond the pre-training window, to our knowledge the first such mechanism for PFNs. In our experiments it beats a recency baseline that gets to keep all its raw data, and we show where it still falls short.

3. Validation against the exact posterior (Section 5): on GP priors we compare the whole pipeline to the true PPD, not just to the teacher, including a regime where the teacher cannot operate at all.

4. A matched-budget comparison with the selectors CRUMB evaluates (uniform subsampling, per-query kNN, MICP, and CRUMB itself), wrapped around the same frozen teacher, plus a covariate-shift stress test (Section 5.1).

2 Background

PFNs approximate the posterior. Let $p(\rho)$ be a prior over tasks and $p(\mathcal{D} \mid \rho)$ a likelihood over data sets. A PFN p_θ is trained by sampling $\mathcal{D} \sim p(\mathcal{D})$, splitting it into a context D_{tot} and queries (x, y), and minimizing $\mathbb{E}[-\log p_\theta(y \mid x, D_{\text{tot}})]$. Müller et al. (2022) show this equals the expected KL from the true posterior predictive to the model plus a constant, so the minimizer is

$$p(y \mid x, D_{\text{tot}}) \;=\; \int p(y \mid x, \rho)\; p(\rho \mid D_{\text{tot}})\, d\rho. \tag{1}$$

For regression, p_θ outputs a *bar distribution*: a piecewise-constant density over B buckets, with borders fixed from prior quantiles (Müller et al., 2022).

The window problem. Let N be the largest context length seen in training. For $|D_{\text{tot}}| > N$, a PFN faces both a cost problem ($O(|D_{\text{tot}}|^2)$ attention) and a distribution-shift problem: it has never seen a context this size. The usual fix, subsampling D_{tot}, throws away likelihood terms in Eq. (1) and provably widens the posterior. Section 5 measures how much this costs.

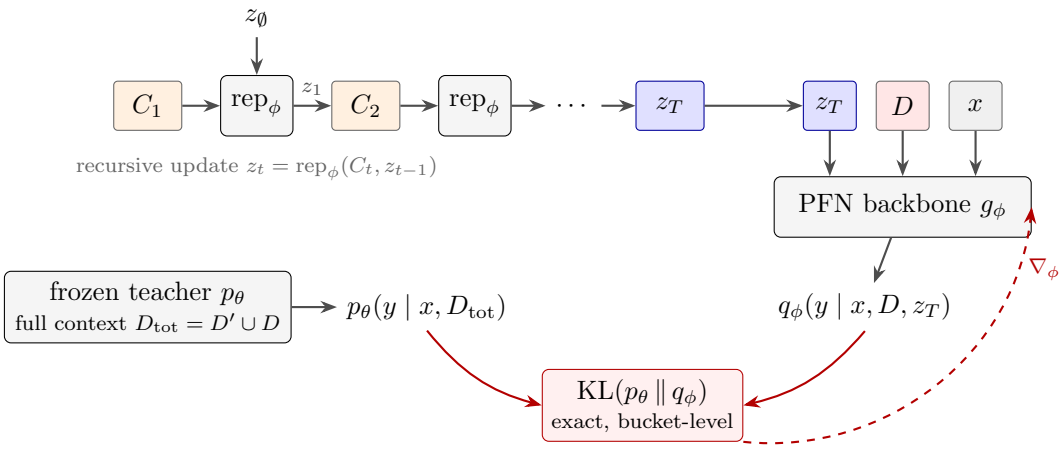


Figure 1: **LongRange-PFN.** D' streams through the summarizer in chunks; the recursive update $z_t = \text{rep}_\phi(C_t, z_{t-1})$ keeps an $O(k)$ state (left). The PFN backbone predicts from $[\, z_T \mid D \mid x \,]$, treating summary tokens as context items (right). Training minimizes the exact bucket-level KL to a frozen full-context teacher (bottom); the teacher is never updated and no ground-truth posterior is needed.

3 LongRange-PFN

3.1 Summary-conditioned prediction

Split $D_{\text{tot}} = D' \cup D$ with $|D'| \gg |D|$. LongRange-PFN has two parts sharing one embedding space:

$$z = \text{rep}_\phi(D') \in \mathbb{R}^{k \times e} \qquad \text{(summarizer)}, \tag{2}$$

$$q_\phi(y \mid x) = g_\phi(x, D, z) \qquad \text{(summary-conditioned PFN)}. \tag{3}$$

The summarizer embeds each pair $(x_i, y_i) \in D'$ with the PFN's own encoders, appends k learned query tokens, and runs PFN-style attention where the queries attend to the data tokens; the resulting query embeddings are z. The predictor is a standard PFN backbone, initialized from the teacher, run on the sequence

$$[\, z_1 \dots z_k \mid D \mid x_{\text{test}} \,], \qquad \text{single_eval_pos} = k + |D|,$$

so summary tokens sit in the context and are attended to just like real samples. When $D_{\text{tot}} = D$ (everything fits), z is a learned null summary $z_\emptyset$ and the model reduces to a plain PFN.

3.2 Predictive KL distillation

Let p_θ be a frozen teacher PFN trained on full contexts. We train ϕ (summarizer and backbone jointly) to minimize

$$\mathcal{L}(\phi) = \mathbb{E}_{D_{\text{tot}} \sim p(\mathcal{D}),\, x}\, \text{KL}\big(\, p_\theta(y \mid x, D' \cup D) \;\big\|\; q_\phi(y \mid x, D, \text{rep}_\phi(D'))\,\big)\,, \tag{4}$$

with $|D|$ drawn from a mixture of small and near-window sizes (Section 5) and, with probability 0.1, $D' = \emptyset$. This is knowledge distillation (Hinton et al., 2015) at the level of predictive distributions (Korattikara et al., 2015; Malinin et al., 2020), but exact: teacher and student share bucket borders, so bar densities over the same borders are supported on disjoint intervals and the KL collapses

Algorithm 1: Recursive KL distillation of LongRange-PFN

Input: frozen teacher p_θ; prior $p(\mathcal{D})$; window N; bucket borders
initialize backbone of q_ϕ from θ; summarizer encoders from θ;
for *each step* **do**
  sample $\mathcal{D} \sim p(\mathcal{D})$ with N context and M query points;
  sample $|D|$ (or degenerate case $D' = \emptyset$); split $D_{\text{tot}} = D' \cup D$; sample $T \in \{1, 2, 3\}$;
  $z \leftarrow z_\emptyset$; **for** $t = 1, \ldots, T$ **do**
    $z \leftarrow \text{rep}_\phi(C_t, z)$
  $\mathcal{L} \leftarrow \frac{1}{M} \sum_x \text{KL}(p_\theta(y \mid x, D_{\text{tot}}) \,\|\, q_\phi(y \mid x, D, z))$; `// exact, bucket-level`
  update ϕ by SGD on $\mathcal{L}$;

to a categorical KL over bucket probabilities, $\sum_b p_b \log(p_b/q_b)$, from the two logit vectors directly. Being a forward KL, it also penalizes the student most where it misses teacher mass, pushing errors toward over-coverage rather than overconfidence.

A small value of Eq. (4) does not by itself bound the gap to the true posterior in Eq. (1): the student can only be as good as the teacher, and KL has no triangle inequality. Section 5 addresses this by comparing against the exact GP posterior, not just the teacher.

A fixed-size z works because conditioning only matters through the posterior over the task:

$$p(y \mid x, D', D) = \int p(y \mid x, \rho)\, p(\rho \mid D', D)\, d\rho, \tag{5}$$

so z only needs to carry what D' contributes to $p(\rho \mid \cdot)$, a learned approximate sufficient statistic (Chen et al., 2021), read by the predictor as a prior that D then updates. This is exactly finite-dimensional for exponential-family tasks; for a GP it is not (the true statistic is the whole posterior process), but its effective dimension shrinks with the kernel's spectral decay, which is why a small k works in practice while still being a real bottleneck (Section 6).

3.3 Recursive summaries: streaming past the window

We make rep_ϕ recursive. Split D' into chunks $C_1, \ldots, C_T$ and set

$$z_0 = z_\emptyset, \qquad z_t = \text{rep}_\phi(C_t,\, z_{t-1}), \qquad t = 1, \ldots, T, \tag{6}$$

where the tokens of z_{t-1} are prepended to the chunk's own tokens inside the summarizer. The final z_T conditions the predictor as before. This mirrors Bayesian filtering: z_t plays the role of the running posterior $p(\rho \mid C_{1:t})$, updated with $O((|C_t| + k)^2)$ attention per chunk and $O(k)$ persistent state. Distillation only needs D_{tot} within the teacher's window, but inference can keep folding in new chunks indefinitely, so the model runs past whatever window it was trained on. Every prefix z_t is also a usable prediction on its own, so the model can answer at any point in the stream, and raw data can be thrown away once it is folded into z, useful for memory, and necessary in settings where the data cannot be kept around at all. During distillation we sample $T \in \{1, 2, 3\}$ uniformly, chunking D' evenly, so one set of weights covers both single-shot and streaming use. Algorithm 1 summarizes training.

3.4 A certificate for information flowing through the summary

An ablation can show that removing z hurts, but we can say something stronger. Since expected NLL is minimized by the true conditional distribution, there is a hard lower bound for any method

Table 1: Positioning. *Amortized*: compression is a single forward pass at test time, no per-dataset optimization. *Latent*: summary lives in embedding space, not data space. *PPD-distilled*: trained against a frozen Bayesian teacher's predictive distribution. *Recursive*: constant-size state updated over a stream. *Beyond window*: demonstrated on more data than the backbone's pre-training window.

Method	Amortized	Latent	PPD-distilled	Recursive	Beyond window
TuneTables (Feuer et al., 2024)	✗	✓	✗	✗	✓
ICD (Ma et al., 2024)	✗	✗	✗	✗	✓
LoCalPFN (Thomas et al., 2024)	✓	✗	✗	✗	✓
MICP (Xu et al., 2024)	✓	✗	✗	✗	✓
CRUMB (Heredge et al., 2026)	✗	✗	✗	✗	✓
Gisting / ICAE (Mu et al., 2023; Ge et al., 2024)	✓	✓	✗	✗	✗
AutoCompressors (Chevalier et al., 2023)	✓	✓	✗	✓	✓
Neural Processes (Garnelo et al., 2018)	✓	✓	✗	✗	—
LongRange-PFN (ours)	✓	✓	✓	✓	✓

that only sees D:

Proposition 1. *Let (D_{tot}, x, y) be drawn from the prior and let q be any predictive distribution that depends on the data only through (x, D). Then*

$$\mathbb{E}\left[-\log q(y \mid x, D)\right] \;\geq\; \mathbb{E}\left[-\log p(y \mid x, D)\right],$$

with $p(y \mid x, D)$ the exact posterior predictive given D alone.

Proof. $\mathbb{E}[-\log q] - \mathbb{E}[-\log p] = \mathbb{E}_{x,D}\,\mathrm{KL}(p(\cdot \mid x, D) \,\|\, q(\cdot \mid x, D)) \geq 0$ (Gibbs' inequality), taking expectations over $y \sim p(\cdot \mid x, D)$. □

In our GP setting $p(y \mid x, D)$ has a closed form, so the bound is something we can actually compute: if a model's test NLL falls below it, that model is using information beyond D. For our student, that information can only have arrived through z. Section 5 shows the student crossing this bound, which is a proof that the summary carries usable posterior information, not just a suggestive ablation.

Remark 1. *Under exchangeability, the order of chunks in Eq. (6) should not matter for the target quantity, but the learned update need not be exactly associative. A regularizer enforcing* $\mathrm{rep}(C_2, \mathrm{rep}(C_1, z_\emptyset)) \approx \mathrm{rep}(C_1 \cup C_2, z_\emptyset)$ *would let shards be summarized in parallel and merged; we leave this for future work.*

4 Related work

Scaling the context of PFNs. TuneTables (Feuer et al., 2024) compresses a large dataset into learned soft prompts for a frozen TabPFN, but as a per-dataset optimization: every new dataset needs its own gradient-descent run, whereas $\mathrm{rep}_\phi(D')$ produces z in one forward pass with no retraining. In-context data distillation (Ma et al., 2024) optimizes, per dataset, a small synthetic set of points instead of tokens, so the summary lives in data space and again needs no amortization. LoCalPFN (Thomas et al., 2024) skips compression entirely and retrieves real kNN neighbors as context. MICP (Xu et al., 2024), the retrieval stage of MixturePFN, clusters the training set and routes each test point to a shared support so queries can be batched. Closest in goal is CRUMB (Heredge et al., 2026), which clusters the test queries and, for each cluster, greedily selects a training

subset that matches its distribution by minimizing maximum mean discrepancy, then runs exact PFN inference on that reduced batch. CRUMB is training-free and architecture-agnostic. It keeps real points rather than a latent, and it needs the test batch before it can choose a context, so it carries no state from one batch to the next the way z_t does. Section 5.1 wraps these four selectors around our teacher at a matched token budget.

A second line changes the architecture instead of the context. TabICL and TabICLv2 (Qu et al., 2025, 2026) factor attention into a column stage and a row stage so the cost in the number of rows is lighter than full self-attention. TabFlex (Zeng et al., 2025) replaces that attention with a linear-time variant. Chunked TabPFN (Sergazinov and Yin, 2025) computes exact attention in tiles, so the context is not reduced at all. These models still condition on raw tokens. They are pretrained tabular foundation models, evaluated on benchmarks such as TabArena (Erickson et al., 2025). The summarizer is a different object: a latent trained by KL against a longer-context teacher.

None of the selectors or the architectural variants trains an encoder against the KL to a full-context teacher's predictive distribution, which is what lets LongRange-PFN transfer calibrated uncertainty rather than point accuracy alone.

Context compression in language models. Gisting (Mu et al., 2023) compresses prompts into gist tokens, ICAE (Ge et al., 2024) trains an autoencoder that produces memory slots, AutoCompressors (Chevalier et al., 2023) build summary vectors recursively over segments (the language-model version of Eq. (6)), and recurrent memory transformers (Bulatov et al., 2022) carry a fixed memory across segments. All train on next-token likelihood over observed tokens rather than a posterior predictive target, so there is no exact distribution-level KL and no exact posterior to check the result against.

Distilling predictive distributions. Distribution-level distillation goes back to Hinton et al. (2015); Bayesian dark knowledge (Korattikara et al., 2015) distills an MCMC-averaged posterior into one network, and ensemble distribution distillation (Malinin et al., 2020) does the same for an ensemble. These compress model averaging into weights. LongRange-PFN compresses something different: part of the conditioning data into a latent state, with an exact KL between two bar-distribution PPDs.

Neural processes and sufficient statistics. Neural processes (Garnelo et al., 2018; Kim et al., 2019) also encode a context set into a latent that conditions a decoder, but both parts are trained from scratch with an ELBO or maximum likelihood, usually with a mean-pooling bottleneck. LongRange-PFN instead distills an already-trained PFN: the teacher fixes the target, the loss is the plain KL of Eq. (4), and the summary is read by an unmodified attention backbone. Treating z as an approximate sufficient statistic connects to neural approximate sufficient statistics for simulation-based inference (Chen et al., 2021) and to amortized in-context posterior estimation (Mittal et al., 2025); neither distills a longer-context teacher or uses a recursive update.

5 Experiments

Setup. We build on the official PFN codebase of Müller et al. (2022).[1] The prior is a 1-d GP (RBF kernel, lengthscale 0.1, outputscale 1.0, observation noise 10^{-2}), chosen so the teacher stays close to the exact posterior at this model scale; a shorter lengthscale (0.03) overwhelmed teacher

[1] https://github.com/automl/PFNs

and summary equally in a preliminary run. Between an 80-point window and a 320-point context, the exact posterior improves by ≈ 0.061 nats, the headroom the streaming experiment is measured against. The teacher is a 4-layer PFN (single token per sample, $e = 128$, 0.9M parameters, $B = 100$ buckets), trained for 12,000 steps on windows $|D_{\text{tot}}| \leq 80$ with 40 query points. The student (1.6M parameters, including a 3-layer summarizer with $k = 16$ tokens) is initialized from the teacher and distilled for 15,000 steps with Algorithm 1, sampling $|D|$ from a small regime ($\mathcal{U}\{5, \ldots, 25\}$) and a wide regime ($\mathcal{U}\{26, \ldots, 75\}$, matching a full local window during streaming), with $n_{\text{chunks}} \in \{1, 2, 3\}$, followed by 8,000 fine-tuning steps after widening the model-selection criterion (below). Checkpoints are selected by lowest summed KL on a fixed, held-out validation set re-evaluated every 375 steps: single-shot regimes $|D| \in \{5, 10, 20, 40\}$ plus the null-summary case during the first stage, with multi-chunk regimes ($|D| \in \{10, 40\}$, $T \in \{2, 3\}$) added for the fine-tuning stage (Appendix A). Distillation improves abruptly rather than smoothly, the summary-reading capability forms in a phase transition around step 7,000 (Appendix B), so shorter runs underperform badly. The pipeline runs in under two hours on a single laptop GPU.

Baselines. *teacher-full* sees all of D_{tot} (the distillation target); *teacher-local* sees only D (plain truncation); *GP exact* is the analytic posterior; *no-z* is the student with the summary replaced by the null token, isolating what z actually contributes.

Closing the truncation gap. Table 2 and Figure 5 sweep $|D|$ with $|D_{\text{tot}}| = 80$ fixed. Truncation costs up to 1.38 nats of test NLL relative to the full-context teacher. The summary recovers almost all of it: at $|D| = 10$ the student closes 97% of that gap, and Figure 5a shows the single-shot student tracking the full-context reference closely across the whole sweep, not just at one point. Removing the summary erases the effect completely, the no-z curve sits on top of teacher-local. At every $|D|$ tested, the student's NLL falls below the Proposition 1 bound: by 1.30 nats at $|D| = 5$, narrowing to 0.03 nats at $|D| = 40$ as D becomes informative on its own (Figure 3). No method that only sees D can do this in expectation, so D' is reaching the predictor through z. Figure 6 shows this on a single draw: the truncated PFN is diffuse away from D, while the student sharpens toward the full-context prediction using z alone.

Multi-chunk summarization is worse. Splitting the same D' into T recursive chunks costs 0.019 nats of KL to the teacher at $T = 1$, rising to 0.283 nats at $T = 5$ (Figure 4), growing roughly linearly in T. Our first hypothesis was a model-selection artifact: the original validation set only evaluated $n_{\text{chunks}} = 1$, so nothing rewarded multi-chunk accuracy. We tested it by adding multi-chunk regimes to the selection criterion and fine-tuning for 8,000 more steps. That improved every regime slightly ($T = 5$: $0.30 \to 0.28$ nats; $T = 1$: $0.025 \to 0.019$) but left the T-dependence essentially intact, so selection explains little of the gap. The near-linear growth instead points to approximation error compounding once per recursive application of rep_ϕ, each update re-encodes the previous summary and loses a little, which suggests training with larger T or the associativity regularizer of Section 3.3 as the next levers, rather than more fine-tuning.

Streaming beyond the teacher's window. Table 3 and Figure 2 test $|D_{\text{tot}}|$ up to 320, four times the teacher's window, under a matched budget: the student keeps the 70 most recent points as D and folds everything older into z recursively (chunks ≤ 70), while the teacher uses the 80 most recent points, the best any window-limited PFN can do. Both attend over the same number of tokens per prediction; the only difference is the $O(k)$ latent carrying the older stream. The student's NLL improves monotonically as the stream grows, running five recursive steps at $|D_{\text{tot}}| = 320$ where training only used three, while the window-limited teacher is flat by construction (it cannot see

Table 2: Predictive quality vs. size of the direct context D, with $|D_{\text{tot}}| = 80$ fixed and D' the remainder (3,200 prior data sets, 40 query points each). Left block: NLL of held-out targets (nats, lower is better). Right block: KL from the full-context teacher (nats/point).

	NLL (nats)						KL to teacher-full (nats)		
$\|D\|$	GP exact (full)	teacher full	teacher local	LR-PFN	LR-PFN (3 chunks)	no-z (abl.)	teacher local	LR-PFN	no-z (abl.)
5	-0.788 ±0.002	-0.724 ±0.002	0.658 ±0.006	**-0.698** ±0.003	-0.506 ±0.003	0.720 ±0.008	1.334 ±0.005	**0.022** ±0.000	1.399 ±0.007
10	-0.788 ±0.002	-0.724 ±0.002	0.141 ±0.006	**-0.702** ±0.003	-0.535 ±0.003	0.171 ±0.007	0.830 ±0.005	**0.020** ±0.000	0.864 ±0.006
20	-0.788 ±0.002	-0.724 ±0.002	-0.349 ±0.004	**-0.708** ±0.003	-0.604 ±0.003	-0.345 ±0.005	0.357 ±0.003	**0.016** ±0.000	0.364 ±0.004
40	-0.788 ±0.002	-0.724 ±0.002	-0.614 ±0.003	**-0.714** ±0.002	-0.664 ±0.002	-0.616 ±0.003	0.105 ±0.001	**0.011** ±0.000	0.105 ±0.001

Table 3: Streaming inference beyond the pre-training window: test NLL (nats) as $|D_{\text{tot}}|$ grows, with matched window budgets, student: $|D| = 70$ recent points + recursive z over the older stream (chunks ≤ 70); teacher: the 80 most recent points (1,280 data sets per cell). LongRange-PFN overtakes the window teacher from $|D_{\text{tot}}| = 160$ onward.

$\|D_{\text{tot}}\|$	GP exact (full)	GP exact (window)	teacher (window)	LR-PFN (stream)
80	-0.786 ±0.004	-0.786 ±0.004	-0.724 ±0.004	**-0.722** ±0.004
160	-0.835 ±0.003	-0.787 ±0.003	-0.727 ±0.004	**-0.746** ±0.004
240	-0.854 ±0.003	-0.792 ±0.004	-0.731 ±0.004	**-0.755** ±0.004
320	-0.860 ±0.003	-0.788 ±0.003	-0.727 ±0.004	**-0.754** ±0.003

past 80 points). From $|D_{\text{tot}}| = 160$ onward the student is actually better than the window teacher, and the gap grows with stream length: LongRange-PFN recovers 39% of the headroom between the windowed and full-stream exact posteriors. At $|D_{\text{tot}}| = 80$, where one chunk covers all of D', student and window teacher are statistically tied; the advantage appears once the stream needs more than one chunk, exactly when the recency baseline starts discarding data the summary keeps.

A recency window only works if the most recent raw observations are kept around and re-attended at every prediction. LongRange-PFN does not need that: its state is $k \times e$ floats no matter how long D_{tot} gets, while a window has to grow or slide to cover the same ground. That storage argument is asymptotic rather than immediate, at our scale, 2,048 floats only beats keeping the raw points past $\sim 10^3$ observations, so the saving at $|D_{\text{tot}}| = 320$ is architectural, not yet a real memory win. What the experiment adds is that discarding the raw data costs nothing here in accuracy, which also matters when the data cannot be retained at all, e.g. for privacy or licensing reasons, and that every prefix z_t is a usable state, so the model can answer mid-stream. Two things temper this result: it is measured against a recency window, not against unbounded full-context inference, and Appendix C shows classical full-context methods still ahead of both; and it uses the same checkpoint whose multi-chunk quality we already flagged as under-optimized (Figure 4), so the real ceiling for the recursion is probably higher than Table 3 shows.

Calibration and where the summary helps. Since training uses a forward KL, the student should preserve the teacher's uncertainty rather than sharpen it away. Coverage of central predictive intervals at $|D| = 10$ confirms this: the student's reliability curve sits closer to the diagonal than truncation (which under-covers) or the full-context teacher (mildly conservative). Binning test points by distance to the nearest point of D shows the student's gain over truncation growing with that distance and vanishing near D (Figure 7), the spatial signature of the certificate in Section 3.4, and it also shows how much of the full-context gain is still uncaptured far from D. A linear probe

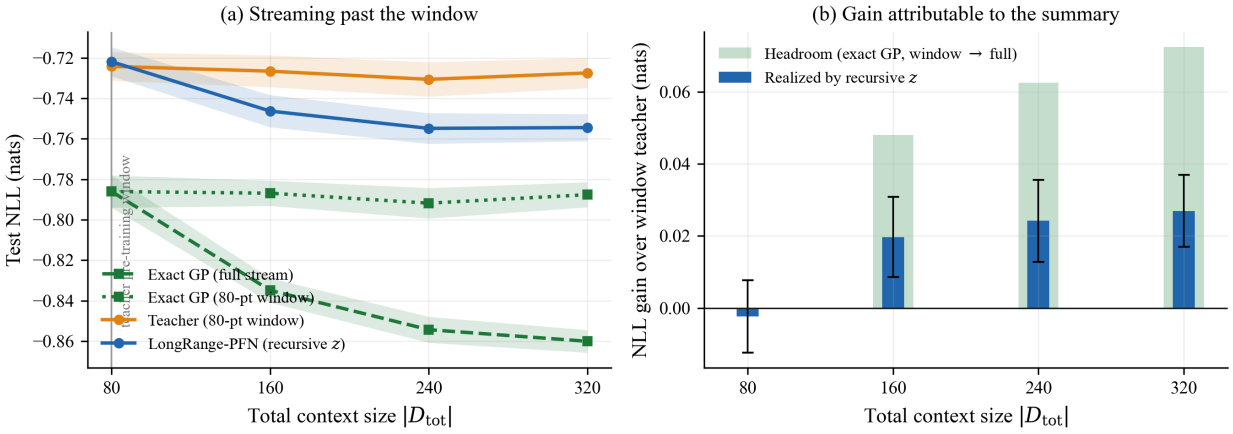


Figure 2: **Streaming beyond the pre-training window** (matched window budgets, mean ± 95% CI over 1,280 data sets per point). (a) Test NLL vs. total context size: the window-limited teacher (orange) is flat by construction, the exact GP on the full stream (green, dashed) improves with evidence, and the recursive student (blue) tracks the teacher at $|D_{\text{tot}}| = 80$ and then surpasses it as the stream grows. (b) The gain over the window teacher realized by the summary (blue bars) against the headroom between the windowed and full exact posteriors (green bars); the recursion recovers roughly a third of it.

of z (Appendix D) finds the full-context predictive mean almost fully linearly decodable from $\text{vec}(z)$ across the whole input domain (held-out $R^2 \approx 0.97$ at every grid location), so z encodes the function itself rather than memorizing individual points.

5.1 Context selection at a matched budget

CRUMB (Heredge et al., 2026) evaluates context selection on TabArena (Erickson et al., 2025), with pretrained TabPFNv2, TabICLv1, and TabICLv2, and reports accuracy and RMSE. Those checkpoints are tabular foundation models. LongRange-PFN is a 1-d GP student distilled from a small teacher, so the measurement that speaks to the summary is to run the same selectors on that teacher and that prior. We reimplement the four selectors from CRUMB's main comparison and wrap them around the frozen teacher. Uniform subsampling draws one shared subset. Per-query kNN gives every test point its own n nearest training points. MICP (Xu et al., 2024) clusters the training pool into $\lceil N/n \rceil$ clusters ($\gamma = 1$, as in CRUMB's Table 1) and routes each test point to the nearest centroid. CRUMB clusters the test set into $K = 8$ groups (their $K = 20$ is for a few hundred queries; with 40 queries, $K = 8$ keeps the same cluster-to-query ratio) and fills each group by the greedy MMD step in their Equation (3), with an RBF kernel whose bandwidth is the median pairwise distance on the standardized training inputs. Standardization uses the training mean and variance, so a shifted test set stays shifted.

Every selector attends n raw tokens. LongRange-PFN attends n raw tokens plus the $k = 16$ summary tokens. In the random-window variant the raw window is the last n points of the pool, a random subset under the prior, and the complement is summarized, in one chunk when it fits and in chunks of 70 otherwise. In the MMD-window variant the raw window is the single subset a $K = 1$ CRUMB run would have kept for the whole test set, and the complement is summarized. KL to the teacher on the full pool is reported only when that pool fits in the teacher's training window.

Table 4 and Figure 8 (left) cover $N = 80$ at budgets $n \in \{8, 16, 40\}$, which are 10%, 20%, and

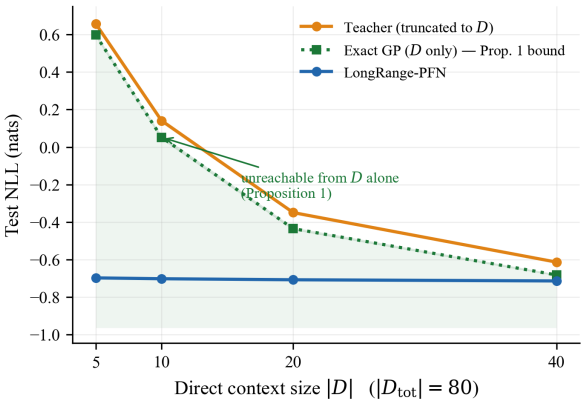


Figure 3: **The Proposition 1 certificate.** The exact GP restricted to D (green, dotted) lower-bounds the expected NLL of any method that sees only D; the shaded region is unreachable without outside information. The student (blue) sits inside it at every tested $|D|$, close to the full-context optimum.

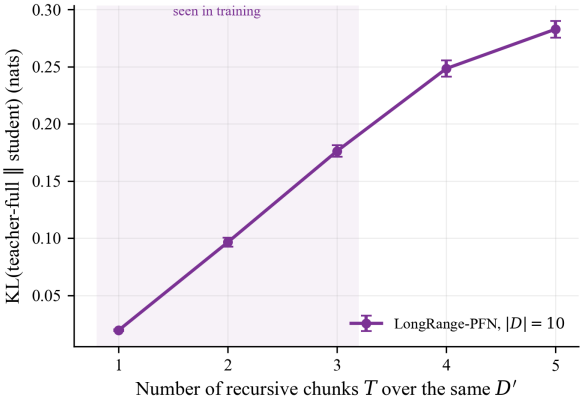


Figure 4: **Recursion cost.** KL to the full-context teacher when the same D' ($|D'| = 70$, $|D| = 10$) is folded in T recursive chunks. Cost rises from 0.019 nats at $T = 1$ to 0.283 nats at $T = 5$; $T \in \{4, 5\}$ were never seen during distillation (shaded: training range). The cost persists after adding multi-chunk regimes to the model-selection criterion and fine-tuning, so it is not a selection artifact; its near-linear growth in T is consistent with error compounding once per recursive step.

50% of the pool. Uniform subsampling is far from the teacher at the two smaller budgets (KL 0.966 at $n = 8$, 0.528 at $n = 16$). At $n = 8$, per-query kNN is the closest of the budget-matched methods (KL 0.066). The random-window student (KL 0.087) and the MMD-window student (KL 0.083) sit between kNN and CRUMB (KL 0.112); the student has lower NLL than CRUMB on 66% of data sets and lower NLL than kNN on 44%. At $n = 16$ the order changes. The student's KL to the full-context teacher is 0.018, against 0.030 for kNN and 0.031 for CRUMB, and its NLL is lower than kNN on 67% of data sets and lower than CRUMB on 70%. Sixteen raw points plus a summary of the other 64 match the teacher more closely than any selector allowed to keep only sixteen real points. At $n = 40$, CRUMB has the smallest KL (0.004) and the test NLL of CRUMB, kNN, and both student variants lie within one standard error of each other, just under the full teacher. With half the pool in the raw window, the summary has less left to add.

Table 5 repeats the comparison at $N = 320$, four times the training window, where a full-context teacher forward is outside the regime the teacher was trained for. At $n = 32$, per-query kNN reaches -0.753, the MMD-window student -0.737, the random-window student -0.732, and CRUMB -0.719, all well below uniform subsampling (-0.576). At $n = 64$, kNN (-0.775) and CRUMB (-0.774) tie, and the student (-0.762) is about 0.01 nats behind. That budget folds the complement into four or five recursive chunks, the regime where each application of rep_ϕ already loses information (Figure 4). A query-aware selection of a window-sized raw context is the stronger predictor there.

Table 6 is the shift experiment from CRUMB's Section 5.4, run on the GP prior. Points are ranked by x. The training pool is the lower half. The test window is $[\tau/2, 0.5 + \tau/2]$. At $\tau = 0$ the MMD-window student is best (-0.749) and the random-window student (-0.738) is ahead of kNN (-0.709) and CRUMB (-0.708). The complement still lies in the test region, so z carries residual evidence the raw window dropped. At $\tau = 1$ every method degrades. kNN, MICP, and

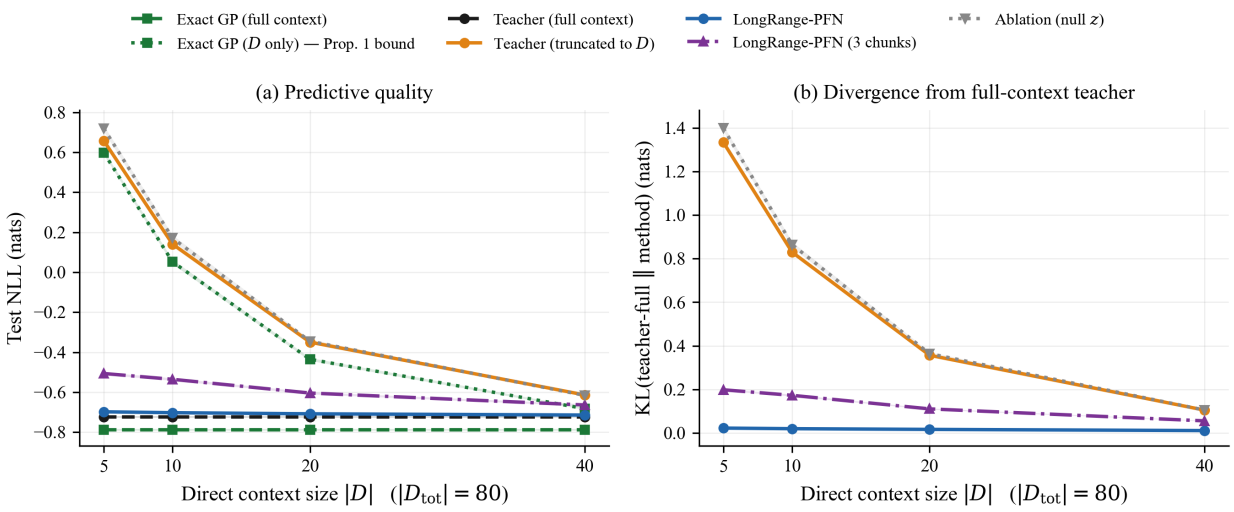


Figure 5: **Closing the truncation gap** (mean ± 95% CI over 3,200 data sets). (a) Test NLL vs. $|D|$: the single-shot latent summary (blue) tracks the full-context reference (black, dashed) almost exactly and stays below the exact-GP-on-D bound (green, dotted); the 3-chunk variant (purple) trails both, consistent with Figure 4. (b) KL to the full-context teacher; the no-z ablation (gray) tracks plain truncation.

CRUMB land together (1.204, 1.210, 1.226). The random-window student rises to 2.575 and the MMD-window student to 2.956, both worse than uniform subsampling (1.323). The summary was trained on exchangeable draws from one region. On a disjoint test set, z encodes points the queries do not resemble, and the student conditions on that state. A selector that withholds those points degrades less.

The split is the practical one. With a very small budget, or with a budget as large as the window on a long pool, keeping real neighbors wins. With a moderate budget and a test distribution that matches the pool, the distilled summary of the discarded points wins. When the test distribution moves and the summary was trained only on exchangeable data, test-side retrieval wins by about a nat. CRUMB still requires the test batch before it can build a context, and it keeps no state across batches. The recursive summary is what still runs when the stream arrives before the queries.

6 Limitations

This is a proof of concept on a 1-d GP prior with a small teacher and exchangeable contexts. Chunk boundaries do not matter here but would under non-stationary or ordered data, where the recursion should be treated as a proper filter. The summary size k is an information bottleneck: a finite k cannot represent posteriors that concentrate on high-dimensional structure, and we have not characterized the rate–distortion trade-off. The distillation ceiling is the teacher's own quality within its window; past that, the recursion is validated only empirically. The multi-chunk cost (Figure 4) is a real limitation of the current recursion: we ruled out the model-selection explanation by widening the validation criterion and fine-tuning, and the near-linear growth in T remained, so each recursive application of rep_ϕ loses information that more training does not recover. A more basic limit shows up in Appendix C: a fitted classical GP or full-context TabPFN still beat bounded-state inference here, simply because exact inference on this prior is already cheap. A real advantage over such

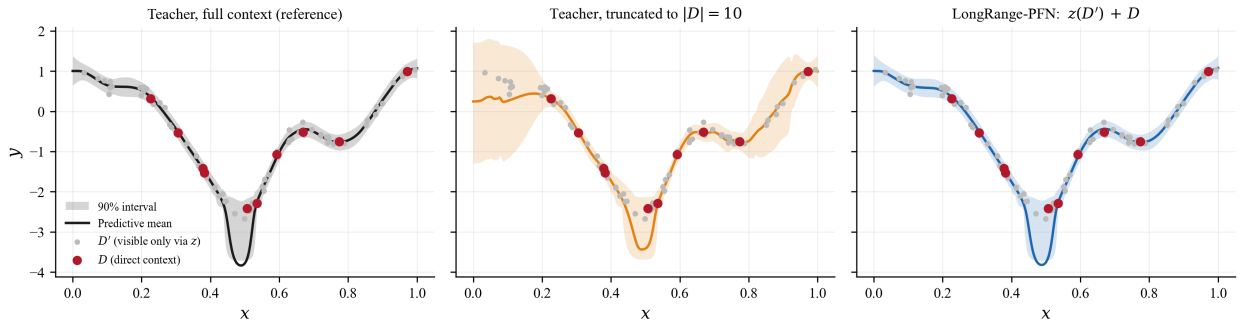


Figure 6: **Qualitative behaviour** on one GP draw ($|D| = 10$, red points; D', gray points, visible to the student only through z). Predictive mean and 90% interval. The truncated PFN (middle) is overdispersed away from D; LongRange-PFN (right) sharpens toward the full-context predictive (left) using the latent summary alone.

Table 4: Matched raw-token budget inside the teacher's window ($N = 80$, 128 data sets, 40 queries). Selectors run the frozen teacher on n real points. LongRange-PFN adds a summary of the complement. NLL is the bar-distribution negative log-likelihood of the held-out targets; KL is to the teacher that sees all 80 points. Bold marks the lowest mean among methods that are limited to budget n. Mean ± s.e.

	NLL (nats)			KL to teacher-full		
Method	$n = 8$	$n = 16$	$n = 40$	$n = 8$	$n = 16$	$n = 40$
Teacher, full context	-0.720 ±0.012	-0.720 ±0.012	-0.720 ±0.012	0.000 ±0.000	0.000 ±0.000	0.000 ±0.000
Uniform	0.299 ±0.026	-0.168 ±0.026	-0.619 ±0.013	0.966 ±0.024	0.528 ±0.021	0.099 ±0.006
Per-query kNN	**-0.646 ±0.013**	-0.685 ±0.012	-0.708 ±0.012	**0.066 ±0.002**	0.030 ±0.001	0.007 ±0.001
MICP	-0.577 ±0.012	-0.634 ±0.013	-0.690 ±0.012	0.126 ±0.003	0.079 ±0.004	0.022 ±0.002
CRUMB	-0.599 ±0.012	-0.683 ±0.012	**-0.713 ±0.012**	0.112 ±0.003	0.031 ±0.001	**0.004 ±0.000**
LongRange-PFN	-0.630 ±0.012	**-0.706 ±0.012**	-0.710 ±0.012	0.087 ±0.005	**0.018 ±0.001**	0.011 ±0.001
LongRange-PFN + MMD window	-0.638 ±0.012	-0.703 ±0.012	-0.711 ±0.012	0.083 ±0.006	0.019 ±0.001	0.012 ±0.001

methods needs a prior where exact inference is intractable and a teacher closer to TabPFN's scale, plus query-aware summaries $z(x)$ and the associativity regularizer from Section 3.3. Section 5.1 adds a separate limit: under a disjoint covariate shift the summary, trained on exchangeable data, degrades by more than the selectors, and pairing it with an MMD window makes that worse.

7 Conclusion

LongRange-PFN carries the evidence a PFN needs from out-of-window data in a small set of latent tokens, trained only by distilling a frozen teacher's predictive distribution. On GP priors, the summary provably carries posterior information, it crosses the exact lower bound of Proposition 1 at every context size tested, and the recursive update handles streams four times the training window at constant state while beating a strong recency baseline. At a matched raw-token budget the summary is closer to the full-context teacher than uniform subsampling, kNN, MICP, or CRUMB when $n = 16$; per-query kNN wins at $n = 8$, and under a disjoint covariate shift the selectors degrade less (Section 5.1). Two further gaps remain: multi-chunk summarization still lags single-shot after the validation criterion was widened, and bounded-state inference still trails classical full-context methods on this easy prior. Both are concrete and measurable, which is what the exact-posterior certificate is for.

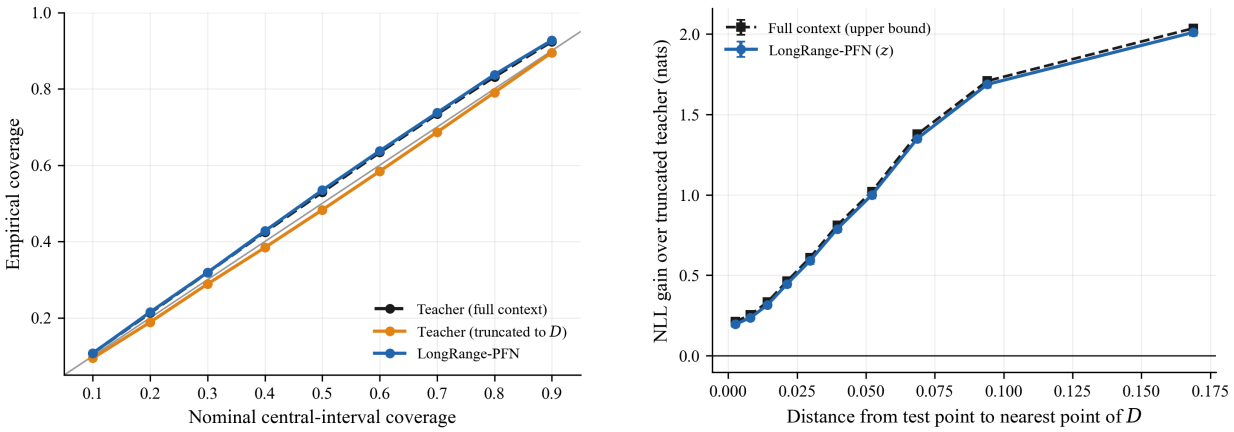


Figure 7: **Diagnostics at $|D| = 10$.** Left: coverage of central predictive intervals; the student (blue) sits closest to the diagonal, truncation (orange) under-covers, and the full-context teacher (black, dashed) is mildly conservative. Right: NLL gain over the truncated teacher, binned by distance from the test point to the nearest point of D (mean ± 95% CI); the gain grows with distance and stays below the full-context ceiling (dashed).

Table 5: Past the teacher's window ($N = 320$, 64 data sets). The teacher is not run on all 320 points. Entries are bar-distribution test NLL. Bold is the lowest mean at that budget. Mean ± s.e.

Method	$n = 32$	$n = 64$
Uniform	-0.576 ±0.019	-0.708 ±0.016
Per-query kNN	**-0.753 ±0.014**	**-0.775 ±0.013**
MICP	-0.716 ±0.015	-0.743 ±0.014
CRUMB	-0.719 ±0.014	-0.774 ±0.013
LongRange-PFN	-0.732 ±0.014	-0.762 ±0.014
LongRange-PFN + MMD window	-0.737 ±0.013	-0.759 ±0.013

Reproducibility statement

All code (built on the official PFN repository), training scripts, seeds and configurations, and the scripts that regenerate every number, table and figure in this paper from `results/metrics.json` and `results/context_selection.json` are in the accompanying repository (`long_range_pfn/`, `paper/`).

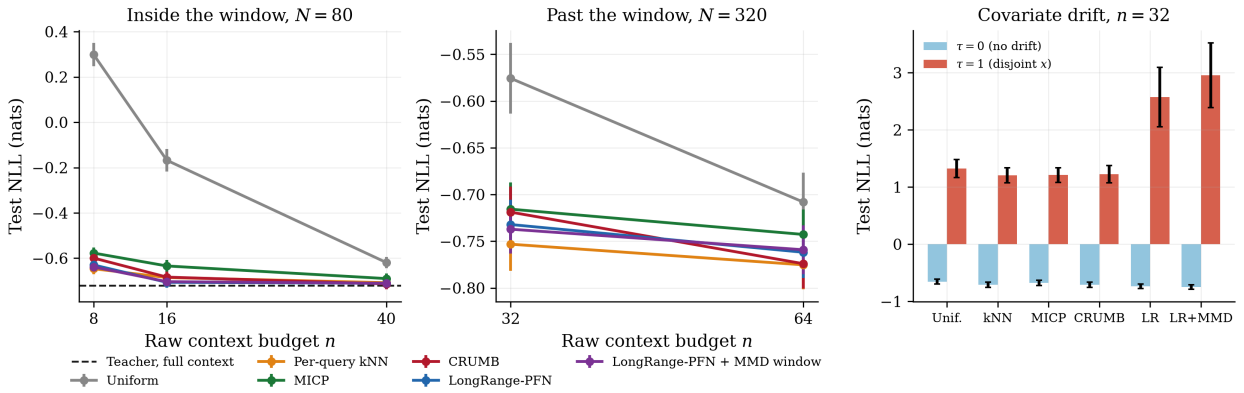


Figure 8: **Selectors versus the latent summary** (mean ± 95% CI). Left: inside the window, uniform subsampling stays far above the full-context teacher until the budget is large, and the student meets that teacher by $n = 16$. Middle: with $N = 320$, per-query kNN is the lowest curve at both budgets. Right: when the test inputs move to a disjoint half of x ($\tau = 1$), the two summary variants degrade past uniform subsampling, while kNN, MICP, and CRUMB stay together.

Table 6: Covariate drift along x, budget $n = 32$, training pool drawn from the lower half of x and capped at 160 points (40 data sets). $\tau = 0$ draws the test window from that same half; $\tau = 1$ draws it from the disjoint upper half. Test NLL, mean ± s.e. Bold is the lowest mean in the column.

Method	$\tau = 0$	$\tau = 1$
Uniform	-0.655 ±0.022	1.323 ±0.080
Per-query kNN	-0.709 ±0.023	**1.204 ±0.066**
MICP	-0.679 ±0.023	1.210 ±0.065
CRUMB	-0.708 ±0.023	1.226 ±0.076
LongRange-PFN	-0.738 ±0.021	2.575 ±0.264
LongRange-PFN + MMD window	**-0.749 ±0.020**	2.956 ±0.289

A Training curves

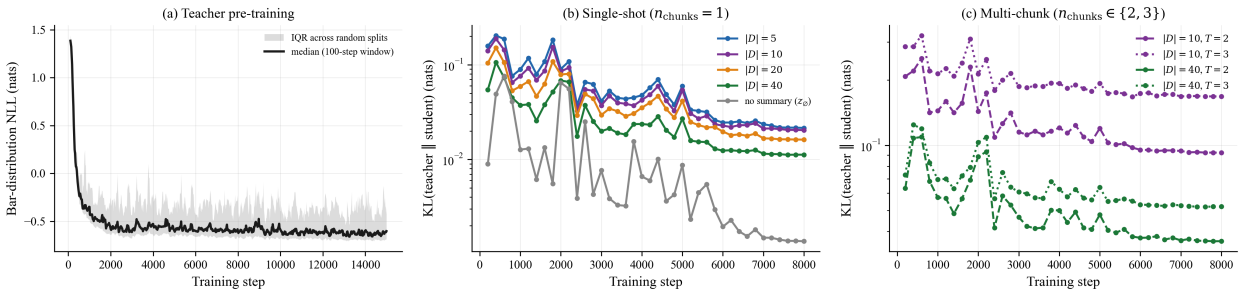


Figure 9: **Optimization traces** behind the reported checkpoint. (a) Teacher pre-training: bar-distribution NLL on prior samples (rolling median, solid; interquartile range, shaded). (b, c) The 8,000-step fine-tuning stage, tracked on the fixed, held-out validation set (resampled once, same batches reused at every evaluation), one line per regime and one panel per chunk count, a single pooled curve over raw training loss would be unreadable, since regimes differ by two orders of magnitude in scale. Single-shot regimes (b) refine at their floor; multi-chunk regimes (c) improve modestly but stay well above it, the gap analyzed in Section 5. The dynamics of the initial 15,000-step stage, including the phase transition, are in Appendix B.

B Training dynamics: a phase transition, not grokking

Distillation does not improve smoothly. Figure 10a shows the fixed-validation KL of the initial 15,000-step distillation stage from Section 5: every regime sits on a plateau for the first $\sim$6,000 steps, drops by one to two orders of magnitude between steps 6,000 and 10,000 (11–49$\times$ depending on regime), then refines slowly. The transition is synchronized, $|D| = 5$, 10 and 20 leave the plateau and reach the floor at the same steps, which points to a single shared mechanism forming at once: the circuit that routes information from the summary tokens into predictions. This mirrors the abrupt formation of induction heads in language models (Olsson et al., 2022), where in-context ability appears in a narrow window of training rather than accruing gradually. The learning-rate schedule is a plain cosine with no restarts, so the window is not a schedule artifact.

It is not grokking. Grokking (Power et al., 2022) is delayed generalization on a memorizable training set: training loss reaches its floor early, held-out loss stays high, and the gap collapses much later. Our distillation samples a fresh prior dataset at every step, so there is no training set to memorize and the training loss is an unbiased estimate of population loss. Figure 10b confirms it: matched-regime training KL and held-out validation KL move together throughout (log–log correlation 0.98), and validation is never worse than training. The jump is a capability transition under online data, not a memorization-to-generalization crossover.

This retroactively explains an earlier failure: a 5,000-step run with the same architecture stopped entirely on the plateau and closed only 12% of the truncation gap, against 97% after the transition. Budgeting distillation past the transition matters more than any other hyperparameter we tuned. Progress-measure analyses of the kind used for induction heads (Nanda et al., 2023), for instance, tracking attention mass from test tokens to summary tokens over training, are the natural tool for identifying what forms inside the window, and we leave this to future work.

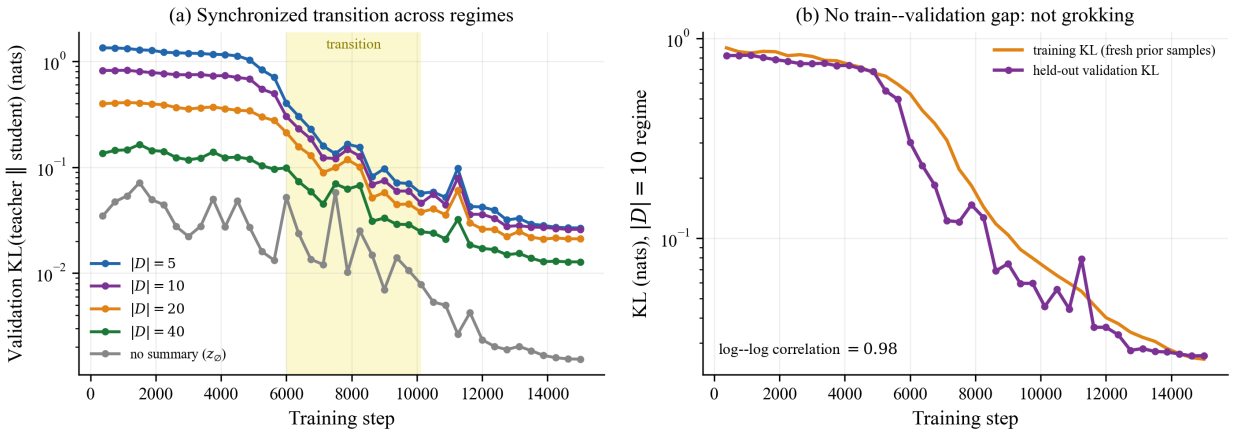


Figure 10: **Training dynamics of distillation** (15,000-step run). (a) Held-out validation KL per regime: a long plateau, then a synchronized 11–49× drop between steps 6,000 and 10,000 (shaded), then slow refinement. (b) The grokking discriminator at $|D| = 10$: training KL on fresh prior samples (matched-regime rolling median) against held-out validation KL. The curves coincide (log–log correlation 0.98); the persistent train–validation gap that defines grokking never forms, as expected with fresh data at every step.

C Defensive baselines at large context

The main text compares against the teacher and the exact GP, asking whether the summary preserves information that truncation destroys. A harder question: why not just give all the data to a method with no window at all? TabPFN, a Gaussian process, and a random forest have no 80-point limit, so here we give them the entire context, up to 8× the teacher's window, and check whether LongRange-PFN, a recent local window plus an $O(k)$ recursive summary, still holds up.

Table 7 covers $N \in \{80, 160, 320, 640\}$, using an earlier teacher/student pair trained with a shorter schedule than Section 5; the conclusion below does not depend on that difference, since it follows from the prior being easy for a matched kernel rather than from any specific checkpoint, but we flag the mismatch and plan to refresh the table. The full-context baselines are TabPFN (Hollmann et al., 2025), a scikit-learn GP with an RBF kernel fit per dataset (unlike the oracle "GP exact", which uses the true generating hyperparameters), kernel ridge regression, k-NN, and a random forest, all given the full N points; LongRange-PFN and the recency-window teacher see only bounded state. We report NLL for the probabilistic methods and RMSE for all.

A fitted RBF-kernel GP ends up close to the exact oracle here, which is not surprising, the data come from an RBF GP, so a matched kernel is close to optimal by construction. Full-context TabPFN is also strong. Both beat the bounded-state methods in NLL by roughly 0.1–0.2 nats. This is an easy point-prediction task for a matched kernel: kernel ridge and the fitted GP get the lowest RMSE, and only k-NN and the random forest are clearly weak, so the setting was never going to show a predictive win over classical methods. That is fine, because the contribution is not a better predictor on GP data, where exact inference is already cheap, it is a bounded-state mechanism for carrying evidence when the full context cannot be stored and re-attended, the case Section 5 argues for. A real predictive win over strong full-data baselines would need a prior where exact inference is intractable and a teacher closer to TabPFN's scale.

Table 7: Large-context defensive baselines over total context size N (up to 8× the teacher's window; 100 datasets per cell, TabPFN on its own smaller sample, noted in text). *Full-context* baselines see all N points; *bounded-state* methods (teacher-window, LongRange-PFN) see only recent points plus, for LongRange-PFN, the $O(k)$ recursive summary. NLL (nats) for probabilistic methods, RMSE for all; mean ± s.e.

NLL (nats, lower is better)				
Method	$N = 80$	$N = 160$	$N = 320$	$N = 640$
Teacher (80-pt window)	-0.693 ±0.016	-0.682 ±0.015	-0.697 ±0.014	-0.716 ±0.014
LongRange-PFN (bounded)	-0.678 ±0.016	-0.662 ±0.014	-0.684 ±0.014	-0.702 ±0.012
GP exact (full, oracle)	-0.789 ±0.012	-0.824 ±0.011	-0.859 ±0.011	-0.893 ±0.010
TabPFN (full context)	-0.632 ±0.024	-0.723 ±0.020	-0.782 ±0.020	-0.756 ±0.024
GP-RBF fitted (full)	-0.771 ±0.014	-0.815 ±0.011	-0.856 ±0.011	-0.891 ±0.011
RMSE (lower is better)				
Method	$N = 80$	$N = 160$	$N = 320$	$N = 640$
Teacher (80-pt window)	0.142 ±0.008	0.154 ±0.009	0.148 ±0.009	0.140 ±0.008
LongRange-PFN (bounded)	0.141 ±0.008	0.153 ±0.008	0.147 ±0.009	0.139 ±0.008
GP exact (full, oracle)	–	–	–	–
TabPFN (full context)	0.120 ±0.004	0.106 ±0.002	0.103 ±0.002	0.101 ±0.002
GP-RBF fitted (full)	0.111 ±0.002	0.106 ±0.001	0.102 ±0.001	0.099 ±0.001
Kernel ridge (full)	0.110 ±0.002	0.106 ±0.001	0.102 ±0.001	0.098 ±0.001
k-NN (full)	0.164 ±0.004	0.125 ±0.002	0.113 ±0.001	0.109 ±0.001
Random forest (full)	0.142 ±0.003	0.127 ±0.002	0.123 ±0.001	0.121 ±0.001

D What does the summary encode?

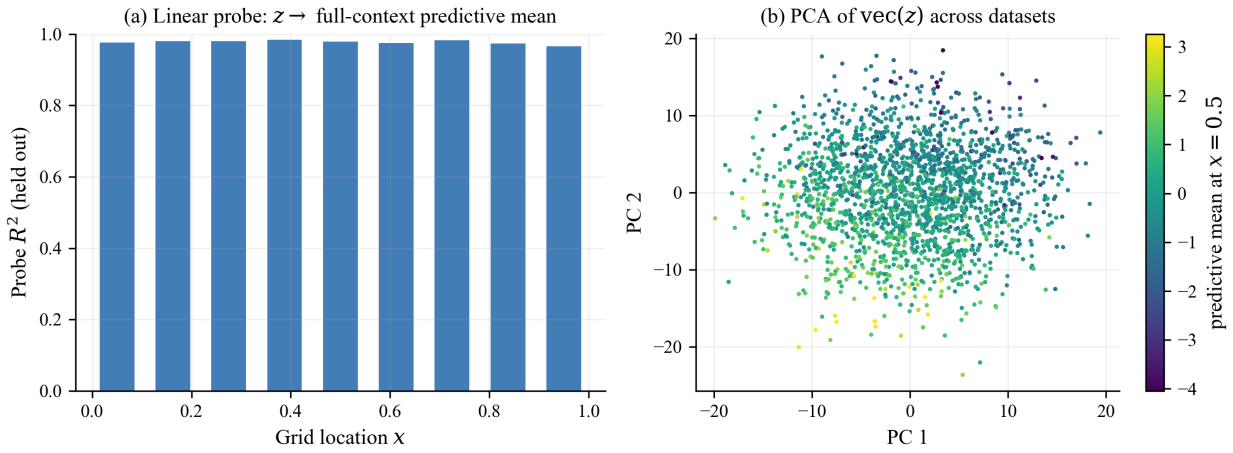


Figure 11: **Probing the summary space** (z computed from all 80 context points; $\approx$ 1,900 held-out datasets). (a) Held-out R^2 of a ridge probe from vec(z) to the full-context teacher's predictive mean at each grid location: after the phase transition of Appendix B, the function is almost fully linearly decodable from the summary everywhere in the domain ($R^2 \approx 0.97$ at every location; a pre-transition checkpoint reached only 0.3–0.9). (b) First two principal components of vec(z), colored by the predictive mean at $x = 0.5$: the summary space is organized by function value along a clear gradient.

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